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Running Symbian OS 7.0, this phone features a 208x320 screen that can display 4,096 colours and is touchscreen capable

The stylus sits in a recessed groove right behind the screen—you can tap away at your documents or Web pages

With 32 MB of memory that is expandable using a Memory Stick interface, the phone also integrates a camera that can capture 640x480, 24-bit images!

breed of cellular phones has graduated into truly smart phones.

In some of the newer phones on the block, we see the convergence of cellular, PDA, as well as imaging technologies. Some even boast of an integrated digital camera, which is capable of capturing images and storing them on the inbuilt memory. For instance, the Nokia 7650 features a brilliant 176x208 pixel colour screen with full PDA functionality. Another phone that integrates a digital camera is the Sony Ericsson T68i, where the Web cam component can be added on to the basic GPRS capable phone—all this with a very small sacrifice in size.

Just rising on the horizon is a smooth blend of PDAs and cellular phones seen in phones such as the Sony Ericsson T800. This phone is for work and for play—you can type in your reports, schedule meetings and play a game. Its keypad folds over to reveal the 208x320 touch screen that can be navigated with a

tiny stylus integrated into the phone. The digital camera is integrated on the rear, much like the Nokia 7650. Pictures are stored on 12 MB of internal memory, which can also be supplemented with a 16 MB Memory Stick. It sports Bluetooth and infrared, and is capable of downloading Java applications such as games. The phone is based on the Symbian 7 operating system.

Another phone that combines funk with innovative features is the Samsung 500, whose flip-open design conceals a full colour screen and has full GPRS support—all in a surprisingly small package.

These new breeds of tablet computers, PDAs and Smart

The tiny Samsung SGH-T100 weighs in at 87 gms and sports a 128x160 flip-top colour screen



The Brains Behind the Beauty

PDAs and Smart Phones will be capable of handling full motion (30 fps) video easily as they would handle mobile gaming and 3G communication. Devices like the HP iPAQ H3950, Pocket LOOX 6 Toshiba Genio E550G and O2 XDA, based on newer processors like Intel's XScale chips are redefining the meaning of mobile computing power.

The XScale processors are available in two variants—the PXA250 (400 MHz, 300 MHz and 200 MHz) and the PXA210 (200 MHz and 133 MHz). They consume up to one-third of the power of their predecessors, the StrongARM processors. The PXA250 is aimed at handheld computers, while the PXA210 is tailored for entry-level handheld cell phones. These processors also incorporate advanced power management features.

By integrating memory and LCD controllers onto the chip itself, they reduce the number of components in the handhelds they are used in. Both these processors come with 64 KB of cache and feature Intel's Integrated Performance Primitives (IPP), which provides built-in functions for signal processing, speech and handwriting recognition, MP3 codecs, MPEG-4 and H.263 video conferencing codes. They also feature Windows CE .Net support.

Besides these processors, there are a slew of other devices, like the Handspring Treo 270 based on the 33 MHz Dragonball VZ processor and the HP Jornada 928, based on the 133 MHz TI OMAP710 processor.

All these enhancements indicate the new super-breed of handhelds and Smart Phones will enable users to run applications that were never thought possible, at speeds never experienced before.

Phones blur the lines between communication and portable computing with full video, audio and animation support. Armed with any of these new devices, you can finally experience true freedom from the desktop. So, ready to romp around the world? Don't forget to take your phone with you!

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